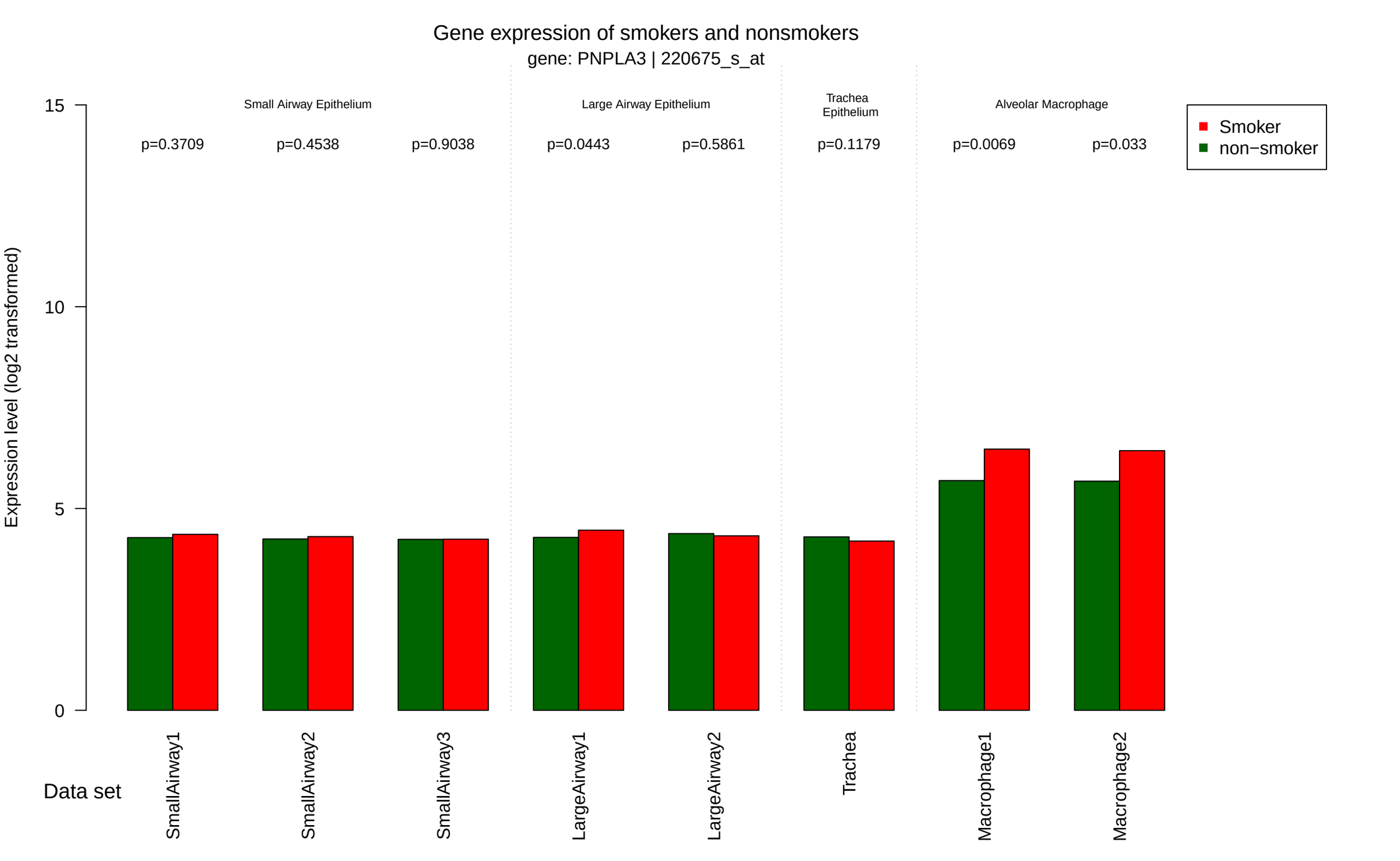
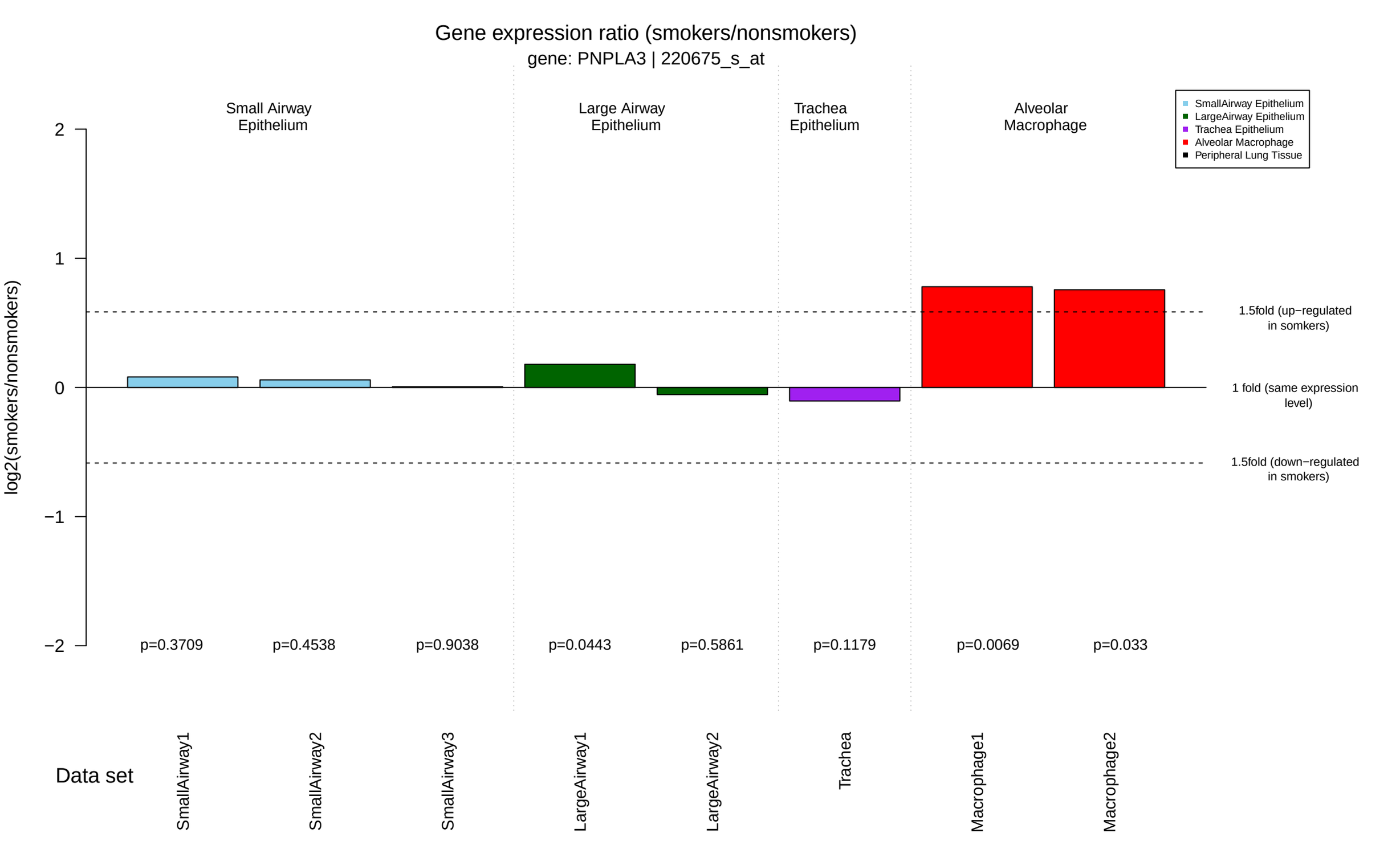
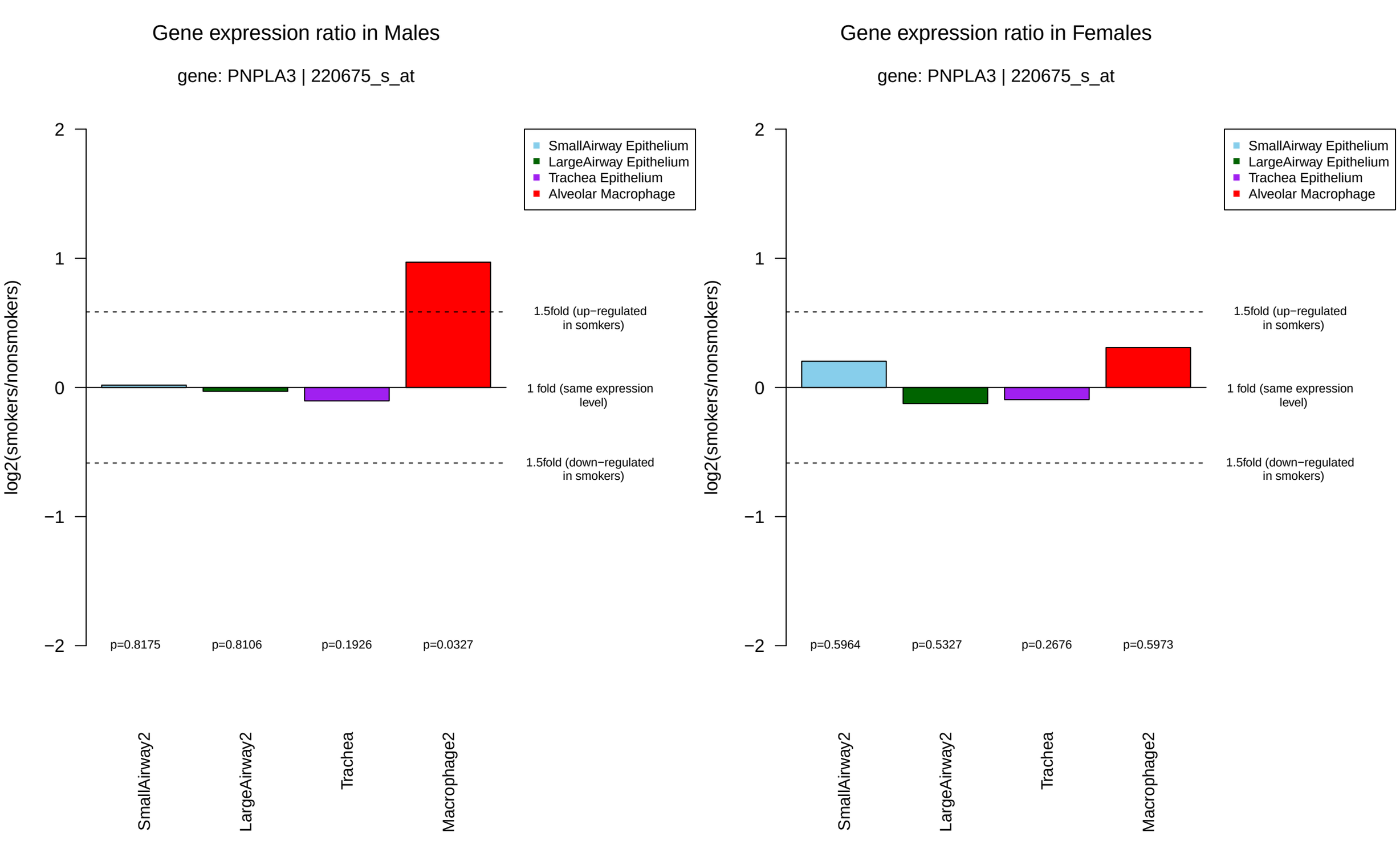
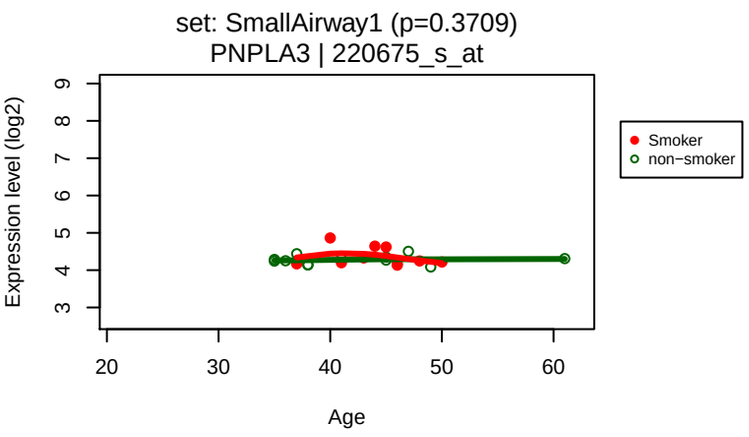
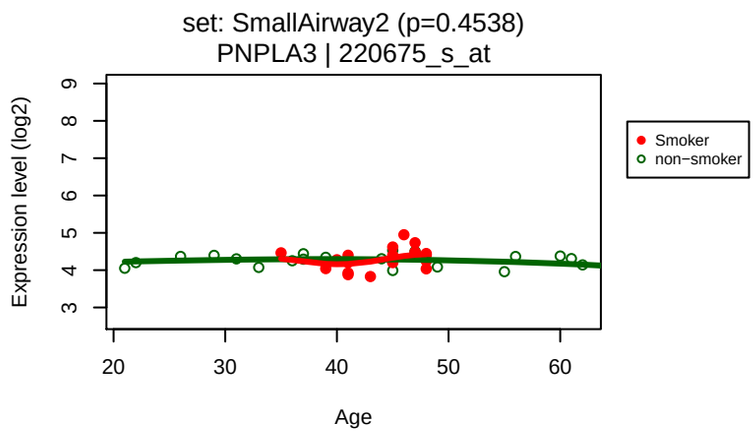
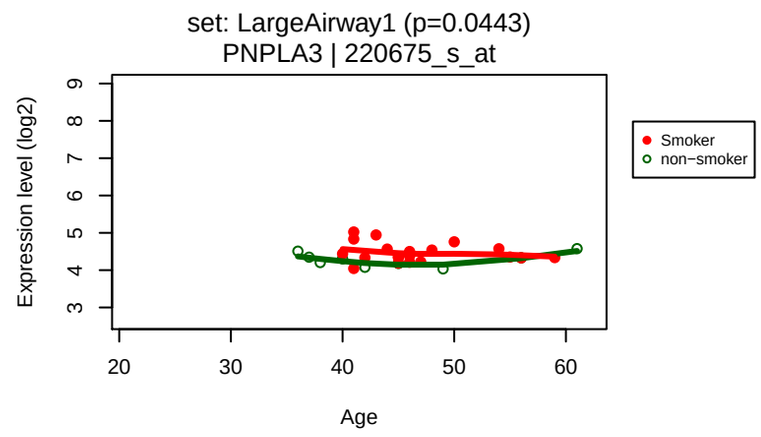

Gene expression of smokers and nonsmokers
gene: PNPLA3 | 220675_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3709
p=0.4538
p=0.9038
p=0.0443
p=0.5861
p=0.1179
p=0.0069
p=0.033
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

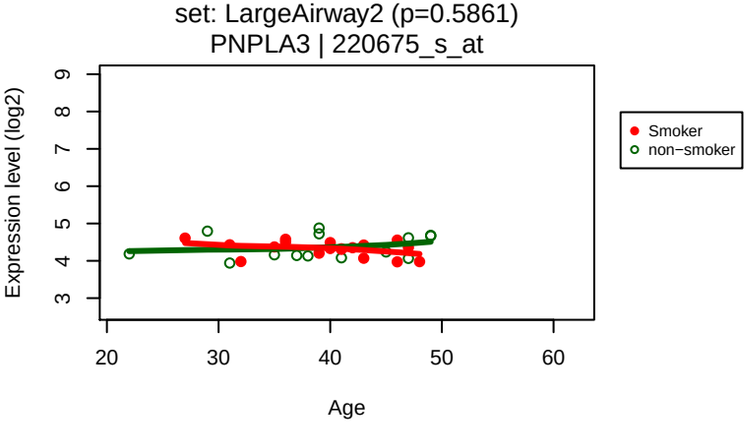

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA3 | 220675_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3709
p=0.4538
p=0.9038
p=0.0443
p=0.5861
p=0.1179
p=0.0069
p=0.033
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

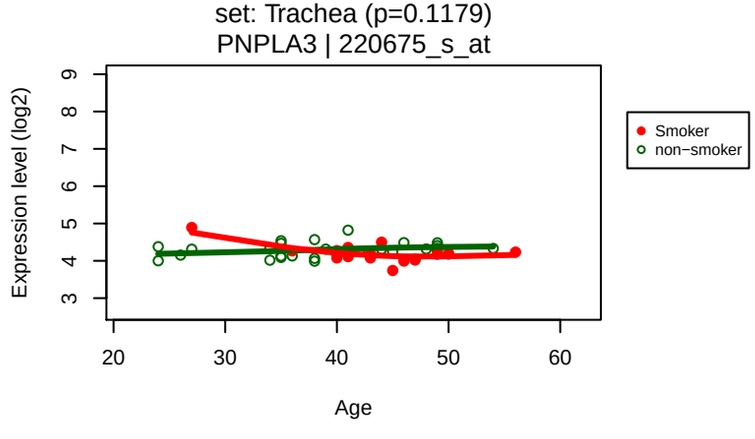

Gene expression ratio in Males
gene: PNPLA3 | 220675_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8175
p=0.8106
p=0.1926
p=0.0327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA3 | 220675_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5964
p=0.5327
p=0.2676
p=0.5973
SmallAirway2
LargeAirway2
Trachea
Macrophage2

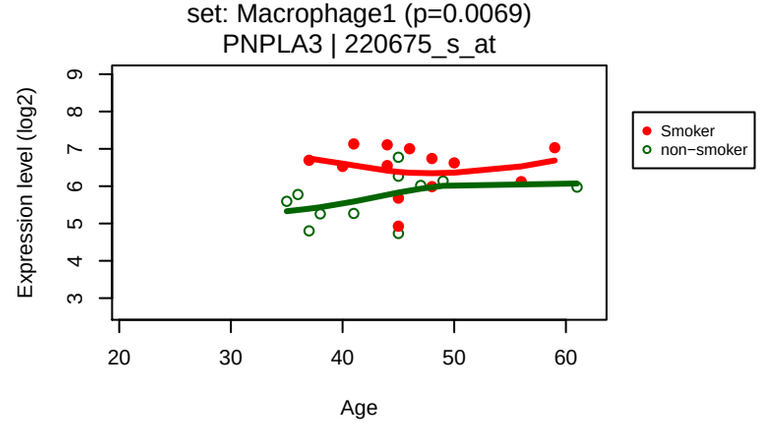

set: SmallAirway1 (p=0.3709)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker

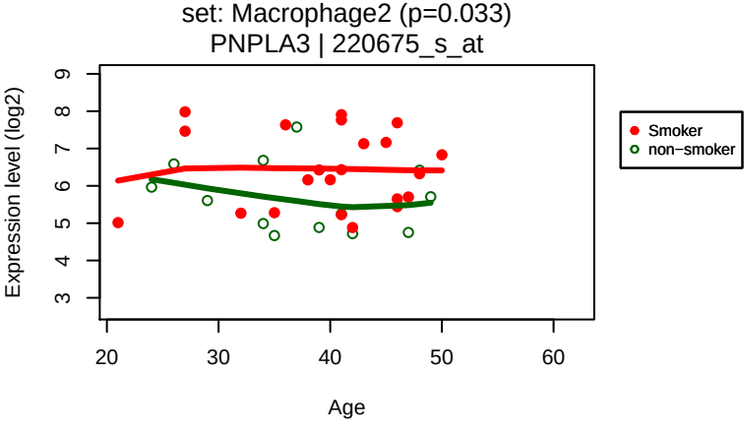

set: SmallAirway2 (p=0.4538)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker

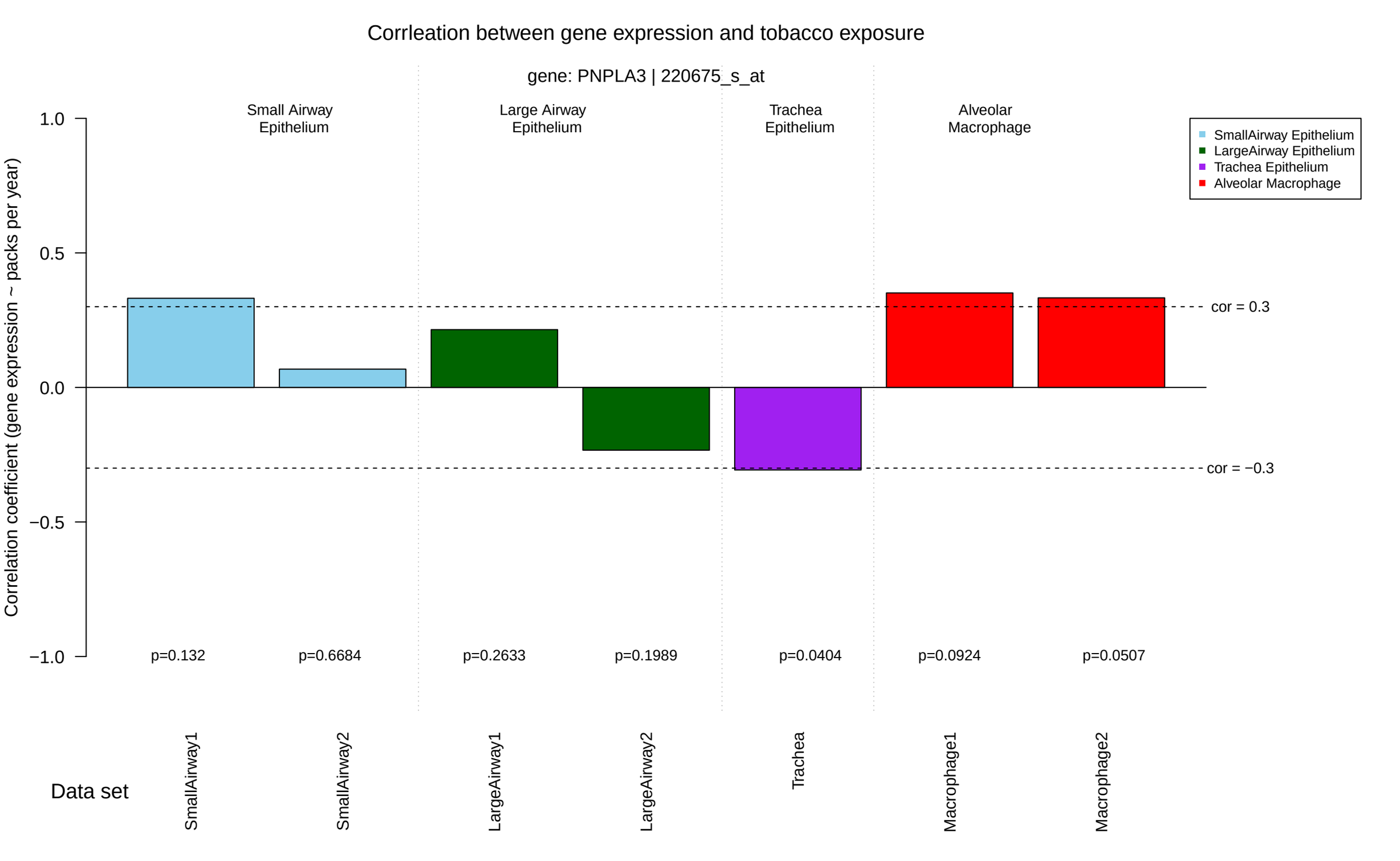

set: LargeAirway1 (p=0.0443)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5861)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1179)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0069)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.033)
PNPLA3 | 220675_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA3 | 220675_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.132
p=0.6684
p=0.2633
p=0.1989
p=0.0404
p=0.0924
p=0.0507
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2